NATIONAL INSTITUTE OF TECHNOLGY PATNA

Department of Computer Science and Engineering

MID SEMESTER EXAMINATION – Feb-March, 2023

B. Tech (Computer Science and Engineering) 4th Semester (Section 1 & Section 2)

CS44114 – Automata Theory and Compiler Design

Full marks:30 Time: 2 hours

Answer *all* questions

Q.no	Question	Marks	CO	BL
1	a) Construct a regular expression for each of the following languages over $\Sigma=\{a,b\}$: i. Language containing the strings of *a*'s and *b*'s such that each string contains at least one *a* and at least one *b*. $(a+b)^*(ab+ba)(a+b)^*$ ii. Language containing the strings of *a*'s and *b*'s such that each string contains even number of *a*'s. $a^* + (a^*ba^*ba^*)^*$ iii. The set of all strings of *a*'s and *b*'s without any combination of double letters. $b(ab)^*$	2.5x3=7.5	CO1, CO2	Creation
	b) If regular expression *R*=b*ab*ab*, describe *L*(*R*) in simple English.	2.5		Application
2	a) Design an NFA that accepts the regular language mentioned in Q)1)a)ii.	04	CO3	Creation
	b) Design an NFA for the language over {0,1} having all strings such that the third symbol from the right end is *0*.	06	CO3	Creation
3	a) Design a DFA equivalent to the NFA generated in Q)2)b	05	CO3	Creation
	b) Design a DFA that accepts the regular language represented by the following regular expression: *R*=(a|b|c)*ccc(a|b|c)*	05	CO3	Creation